Year 9 Science – Booklet 14: Chemistry

Atomic Structure (2.2 Atoms) — ANSWERS

2.2 Atoms – In-text questions

A. An atom is the smallest part of an element that can exist.

B. Gold atoms are heavier than silicon atoms (1 cm^3 of gold has a mass of 19.3 g, versus only 2.33 g for silicon), and gold atoms are also bigger than silicon atoms.

Summary Questions

1. Copy and complete (3 marks)

The smallest part of an element that can exist is called an **atom**. All the atoms of an element are the **same**. The atoms of one element are **different** to the atoms of all other elements.

2. Bronze medal, three elements (2 marks)

Three types of atom – each element (copper, zinc, tin) is made up of only one type of atom, and since three different elements are present, there must be three different types of atom.

3. Visual summary (6 marks QWC)

A summary diagram could show: "Atom = smallest part of an element that can exist", with example spheres of different sizes/colours representing atoms of different elements (e.g. a small silicon atom vs a larger, heavier gold atom), plus key facts: there are 92 naturally-occurring elements, so 92 types of atom; and the properties of an element (e.g. colour, shininess, state) only emerge from very many atoms joined together, not from a single atom on its own.

C2.2 Atoms (workbook)

A. Fill in the gaps

Elements are made up of **atoms**. An atom is the **smallest** part of an element that can exist. All the **atoms** of an element are the same. The atoms of one element are **different** from the atoms of all the other elements. The properties of an element are the properties of **many** atoms joined together.

B. Compare copper, mercury and zinc atoms

1. Mercury atoms have a greater radius than zinc atoms, which have a greater radius than copper atoms (mercury is the biggest, copper the smallest).
2. Mercury atoms are heavier (greater relative mass) than zinc atoms, which are heavier than copper atoms.
3. Copper and zinc atoms are similar in size and mass, but mercury atoms are much bigger and much heavier than both.

C. True/false statements about copper and zinc

a). True statements: **2** (the atoms of copper are all the same as each other) and **5** (a copper wire can conduct electricity, but a single copper atom cannot).

b). Corrected versions of the three false statements:
1 → An atom is the smallest part of an element that can exist.
3 → The atoms of copper are different from the atoms of zinc.
4 → A single atom of zinc does not have the same properties as a piece of zinc wire.

D. Solid copper melting

a). In the solid, the copper atoms are arranged in a regular, tightly-packed pattern, touching each other, and only vibrate on the spot. As it melts, the atoms gain enough energy to break free of this fixed arrangement – they move around and slide past each other in a more random arrangement, though they generally stay close together, rather than being locked in fixed positions.

b). Melting is a change of state that only applies to a large collection of atoms (going from a fixed, regular solid arrangement to a moving, randomly-arranged liquid one). A single atom on its own has no such fixed arrangement to break down and no state (solid/liquid/gas) of its own, so the idea of "melting" doesn't apply to it.

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